1 INTRODUCTION.

A cold reader pages backward until it holds 20 exchanges, at least 8. Agent output with no prompt above it counts as one exchange, so a long tool-heavy turn used up the budget first and the reader opened on half a turn with no prompt. The fill now keeps going, on its own bounded allowance, until the prompt that opened the latest turn is loaded (figure 1).

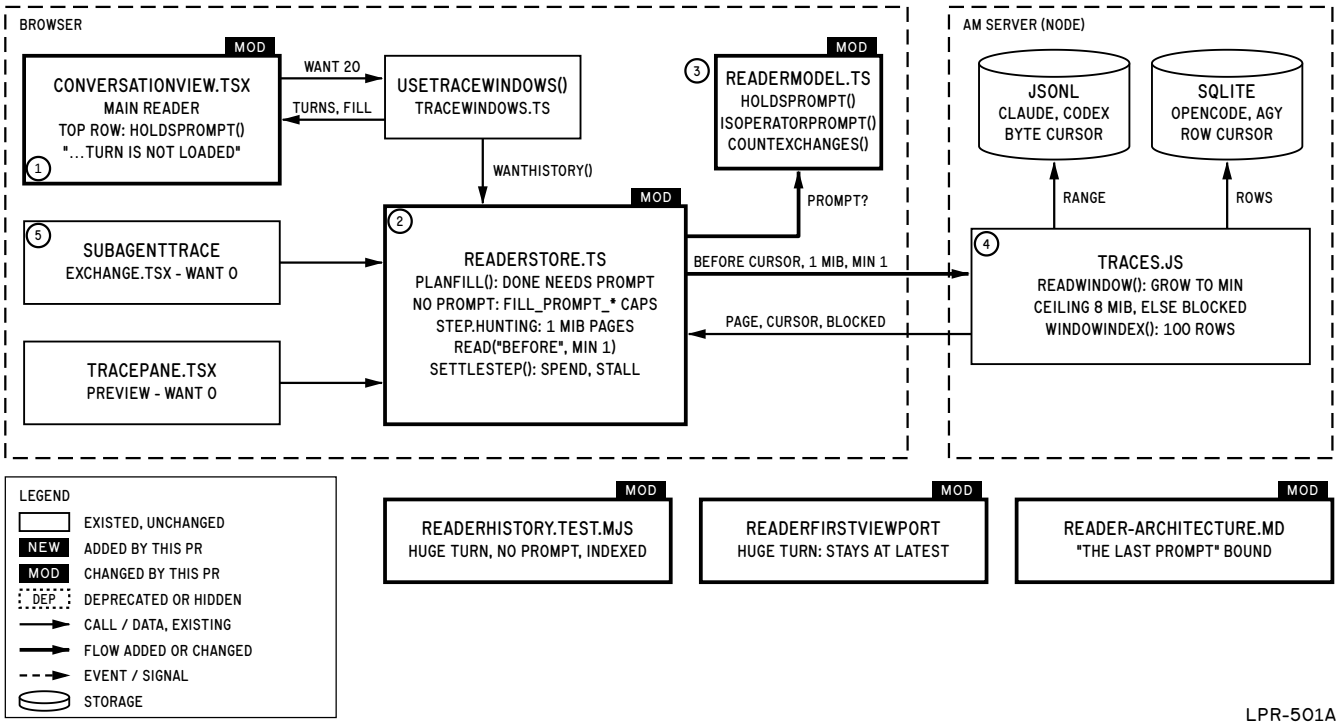


Figure 1. Reader History Fill, Layout and Data Flow

2 HOW IT WORKS.

The main reader asks for 20 exchanges (1). After every accepted read, planFill() decides the next step; it is done only when the count is met and the window holds an operator prompt (2), checked by holdsPrompt(), which ignores still-queued prompts (3). Without one it pages back in 1 MiB windows on the FILL_PROMPT_* caps; the server grows a window to one whole record, up to 8 MiB (4). Child readers and previews ask for nothing and keep their behaviour (5) (figure 2).

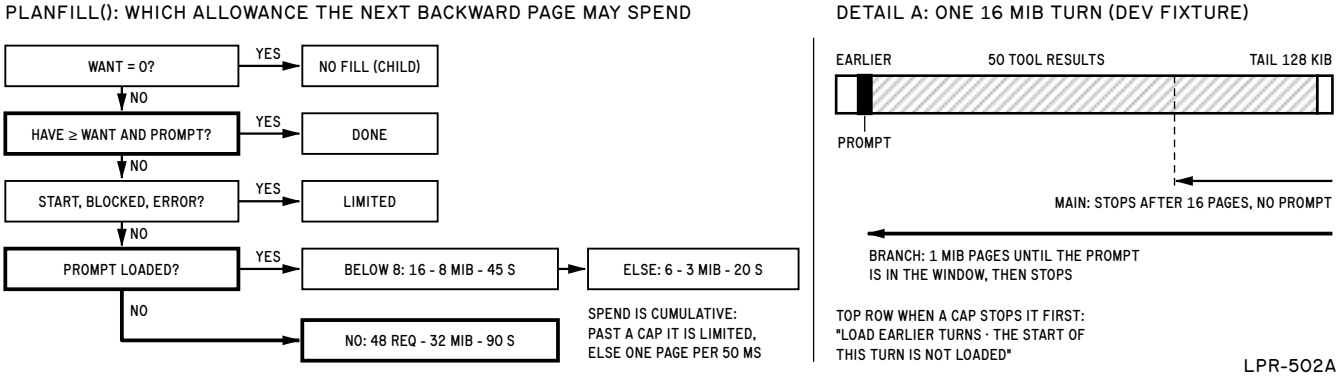


Figure 2. Fill Decision and a Long Turn on the Byte Axis

3 DATA.

Table 1. Last-Prompt Allowance and Measurements

Parameter	Value
FILL_PROMPT_MAX_REQUESTS	48
FILL_PROMPT_MAX_RETAINED_BYTES	32 MiB
FILL_PROMPT_MAX_MS	90 s
FILL_PROMPT_WINDOW_BYTES (byte sources)	1 MiB
Floor allowance, unchanged	16 - 8 MiB - 45 s
Last turn, 94 Claude traces: p50 / p95 / max	0.4 / 3.5 / 9.8 MiB
Indexed source reach (100-row pages)	4,800 rows
Floor and target, unchanged	8 / 20 exchanges

CAUTION

A turn over 32 MiB, or one record over 8 MiB, still opens without its prompt. The reader says so; it does not hide it.

VERIFY LAST PROMPT — **OPERATOR**

1 READER - open a busy agent
  √ prompt above its tool calls
2 VIEW - wait for the fill
  √ still at the latest step
3 TOP ROW - read the label
  √ "Load earlier turns"
4 CHILD ROW - expand
  √ no extra history reads
5 TURN OVER 32 MIB - open
  √ "…turn is not loaded"